# EC-390: Digital Signal Processing

# *LECTURE NO 8*

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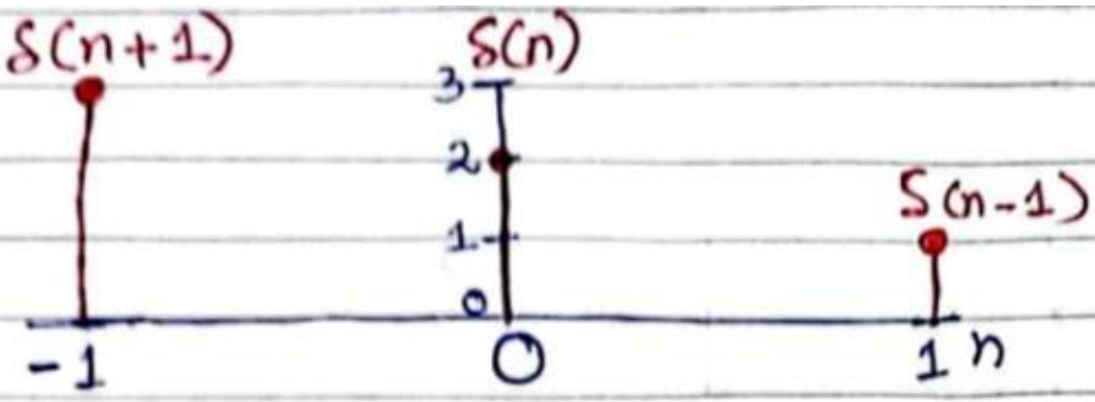
# Organization of Lecture

1. Weighted Shifted Unit Impulse.
2. Impulse Response.
3. Finite Impulse Response.
4. Infinite Impulse Response.
5. Discrete Time Convolution.
6. Properties of Convolution.
7. Auto correlation and Cross correlation.

# 1. Weighted Shifted Unit Impulse

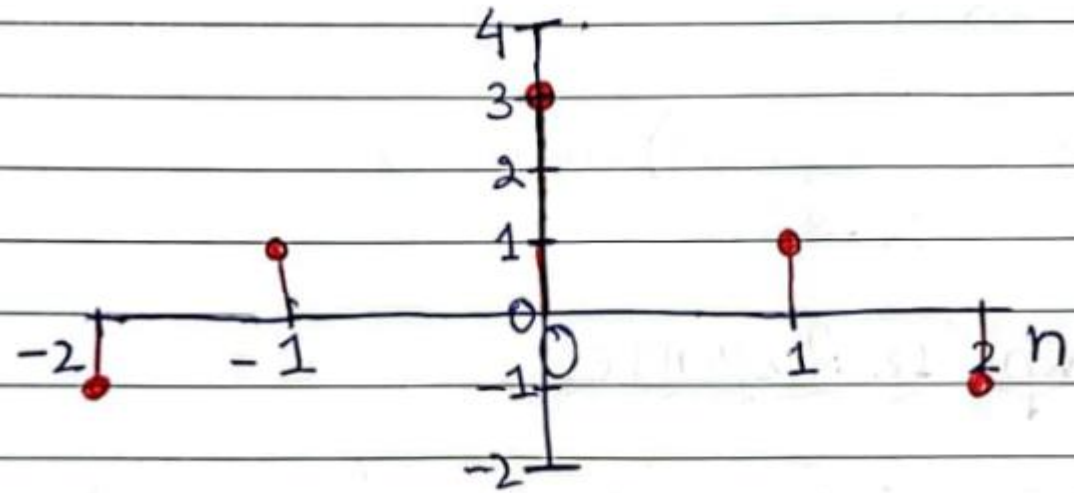



$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k)$$

$$= x(-1)\,\delta(n+1) + x(0)\,\delta(n) + x(1)\,\delta(n-1)$$

$$x(n) = 3\,\delta(n+1) + 2\,\delta(n) + 1\,\delta(n-1)$$

# 1. Weighted Shifted Unit Impulse

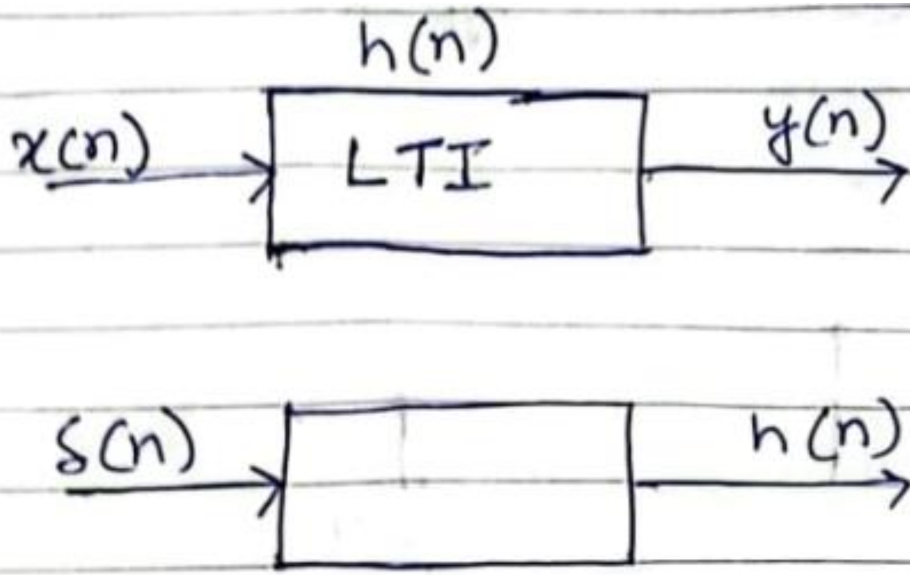


$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k)$$

$$x(n) = x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(0)\delta(n) + x(1)\delta(n-1) + x(2)\delta(n-2)$$

$$x(n) = -1\delta(n+2) + 1\delta(n+1) + 3\delta(n) + 1\delta(n-1) - \delta(n-2)$$

# 2. Impulse Response




* When the input of a system is a Unit Impulse ($\delta(n)$) then the output of the system is called Impulse Response ($h(n)$).

# 3. Finite Impulse Response

* When there are finite no. of samples, the system output is said to be Finite Impulse Response (FIR).

$$y(n) = \sum_{K=-\infty}^{M-1} x(K)\,h(n-K)$$

# 4. Infinite Impulse Response

* When there are infinite no. of samples, the system output is said to be Infinite Impulse Response (IIR)

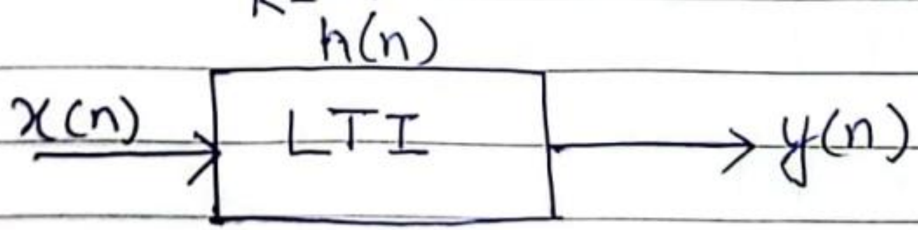
$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \quad \text{—— } (1)$$

# 5. Discrete Time Convolution

* The response $y(n)$ of the system is given by convolution of $x(n)$ and $h(n)$.

· $y(n) = x(n) * h(n)$

$$y(n) = \sum_{K=-\infty}^{\infty} x(K)\,h(n-K)$$




* There are total 4 steps in D.T convolution.

1. Folding
2. Shifting
3. Multiplication
4. Summation.

→ Incase of Graphical Method

# 5. Discrete Time Convolution

* Another (easy) method for D.T convolution is called as Matrix method.
  - → The input sequence $x(n)$ is arranged as a column.
  - → The impulse response $h(n)$ is arranged as a row.

# 5. Discrete Time Convolution

**Example. 1:**

The input $x(n)$ and impulse $h(n)$ are given by

$$x(n) = \{\underset{\uparrow}{1}, 2, 3, 1\}$$

$$h(n) = \{1, \underset{\uparrow}{2}, 1, -1\}$$

Determine the response of the LTI system by:

i- Graphical Method.

ii- Matrix Method.

**Solution:**

**i- Graphical Method**

* $x(n)$ starts at $n=0$, i.e., $n_1 = 0$
* $h(n)$ starts at $n=-1$, i.e., $n_2 = -1$
* The output $y(n)$ starts at $n = n_1 + n_2 = 0 + (-1) = -1$
* Total number of samples in $x(n) = N_x = 4$
* Total number of samples in $h(n) = N_h = 4$
* Total number of output samples $= N = N_x + N_h - 1$

$$N = 4 + 4 - 1$$

$$N = 7$$

# 5. Discrete Time Convolution

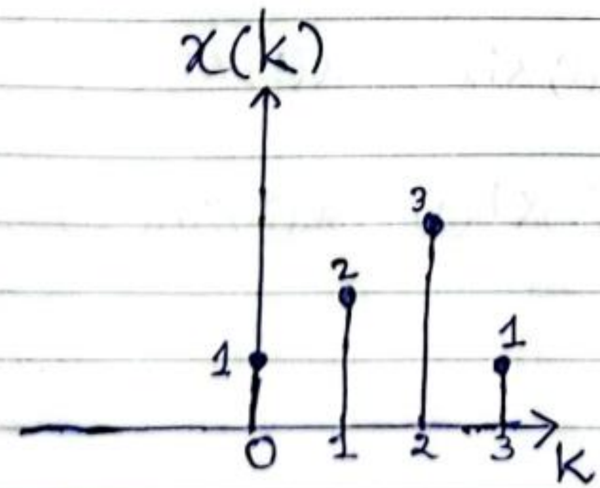
**Step 1:**

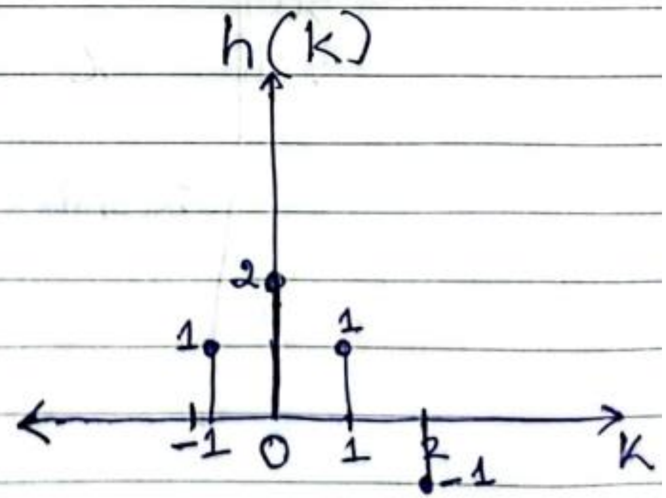
put $n=k$ for both $x(n)$ & $h(n)$ i.e.,

$$x(k) = \{\underset{\uparrow}{1}, 2, 3, 1\}$$

$$h(k) = \{1, \underset{\uparrow}{2}, 1, -1\}$$

Plotting,



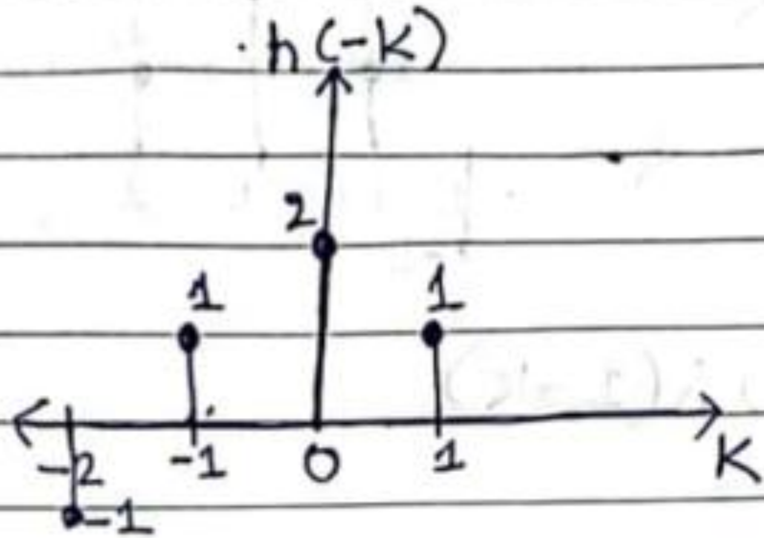



# 5. Discrete Time Convolution

**Step 2**

Apply folding operation on $h(k)$ drawn in step 1. i.e $h(-k)$. Therefore

$$h(-k) = \{-1, 1, \underset{\uparrow}{2}, 1\}$$



# 5. Discrete Time Convolution

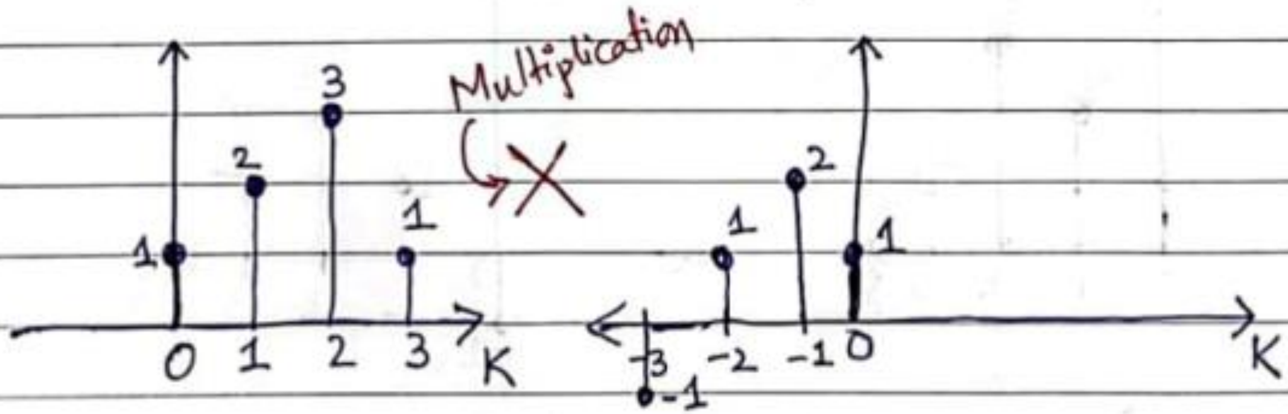
Step 3 and 4

The total no. of output samples will be 7 starting from $-1$. Therefore $n = -1, 0, 1, 2, 3, 4$ and $5$

When $n = -1$:

$$\text{eq } (1) \Rightarrow y(-1) = \sum_{k=-\infty}^{\infty} x(k)\, h(-1-k)$$

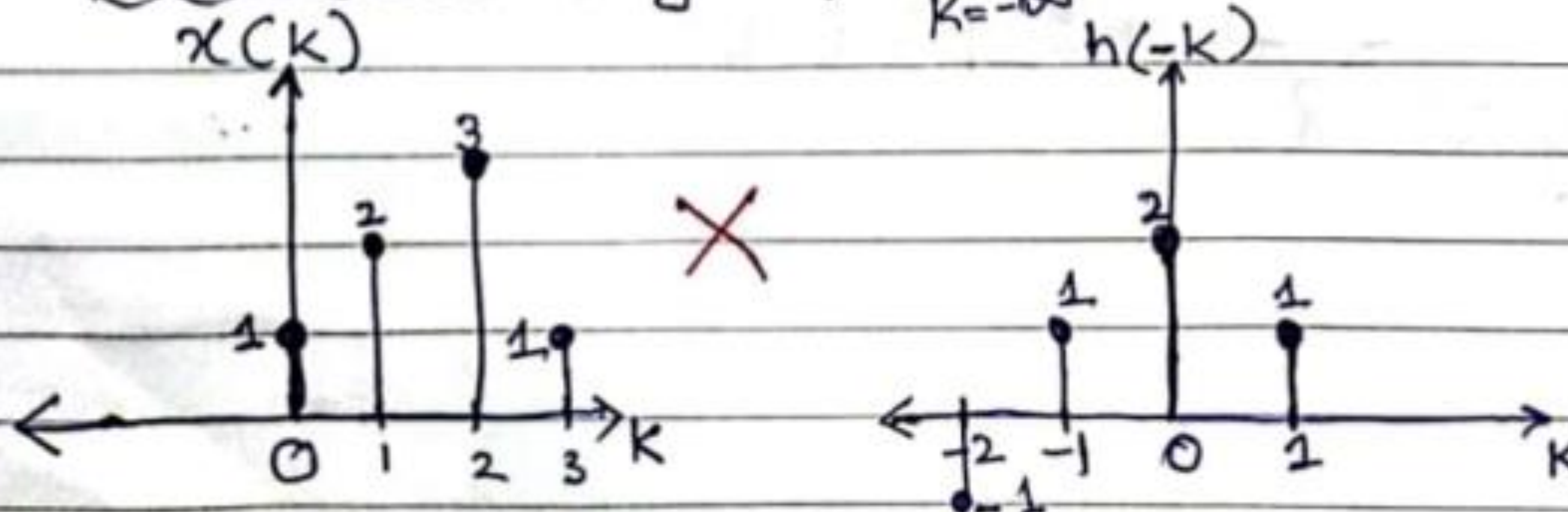
$x(k)$ $\qquad$ $h(-1-k)$ Move $h(-k)$ left by 1 unit




$$y(-1) = x(0) \cdot h(0) = 1 \cdot 1 = 1$$

# 5. Discrete Time Convolution

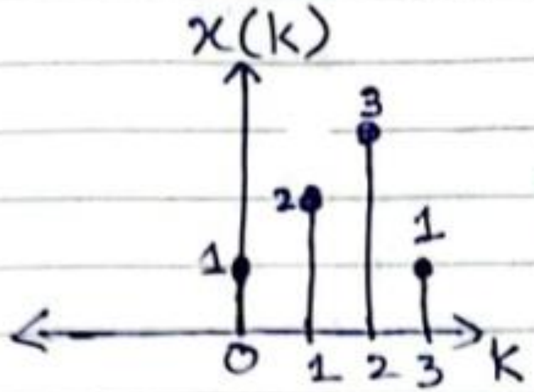
when n = 0 : $y(0) = \sum_{k=-\infty}^{\infty} x(k)\, h(-k)$




$y(0) = x(0).h(0) + x(1)h(1) = (1).(2) + (2)(1) = 2+2 = 4$

# 5. Discrete Time Convolution

When $n = 1$:

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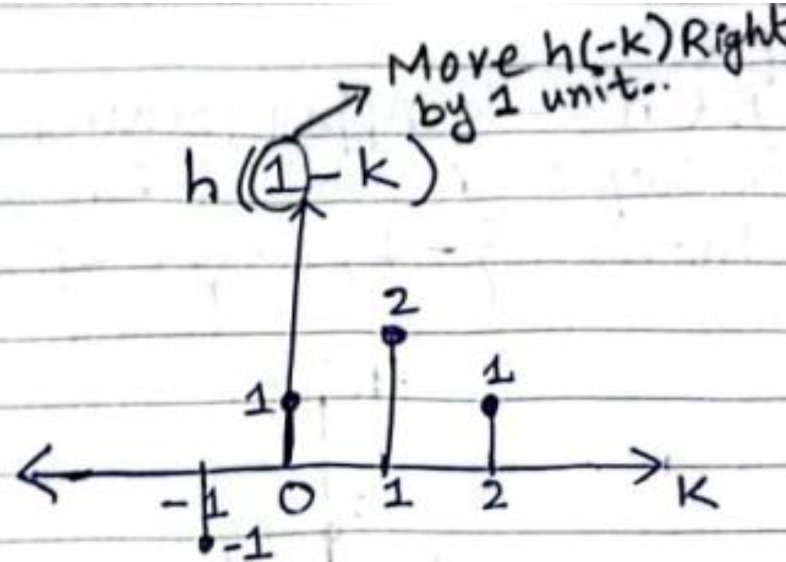



$$y(1) = \sum_{k=-\infty}^{\infty} x(k)\, h(1-k)$$

$$y(1) = x(0)h(0) + x(1)h(1) + x(2)h(2)$$

$$= (1).(1) + (2)(2) + (3)(1)$$

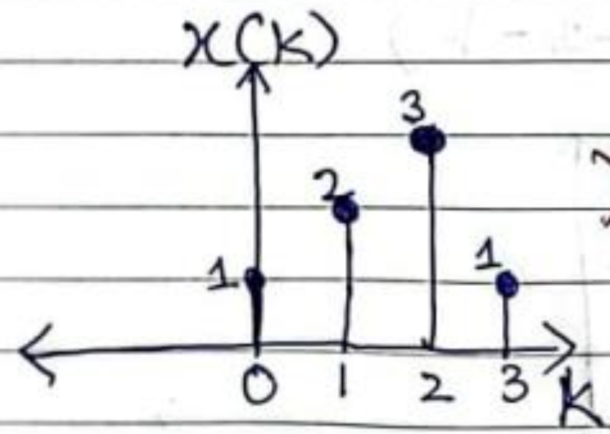
$$= 1 + 4 + 3$$

$$y(1) = 8$$

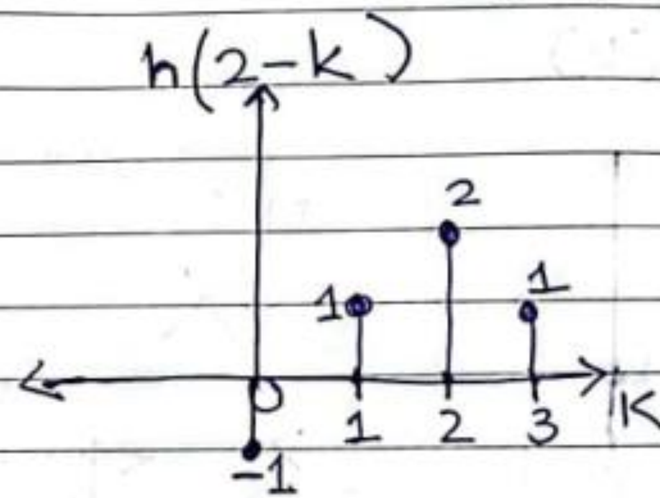
# 5. Discrete Time Convolution

When n = 2

Move h(-k) right by 2 unit




$\times$




$$y(2) = \sum_{k=-\infty}^{\infty} x(k)\, h(2-k)$$

$$= x(0)\cdot h(0) + x(1)\cdot h(1) + x(2)\cdot h(2) + x(3)\cdot h(3)$$

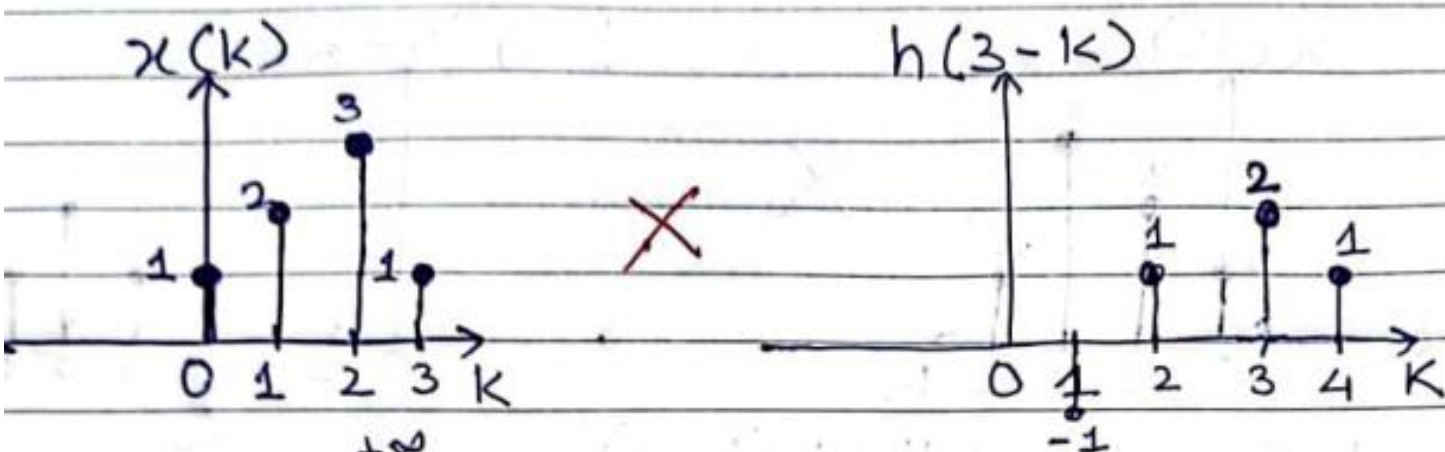
$$= (1)\cdot(-1) + (2)(1) + (3)(2) + (1)(1)$$

$$= -1 + 2 + 6 + 1$$

$$y(2) = 8$$

# 5. Discrete Time Convolution

when $n = 3$:




$$y(3) = \sum_{k=-\infty}^{+\infty} x(k)\, h(3-k)$$

$$= x(1).h(1) + x(2).h(2) + x(3).h(3) + x(4).h(4)$$

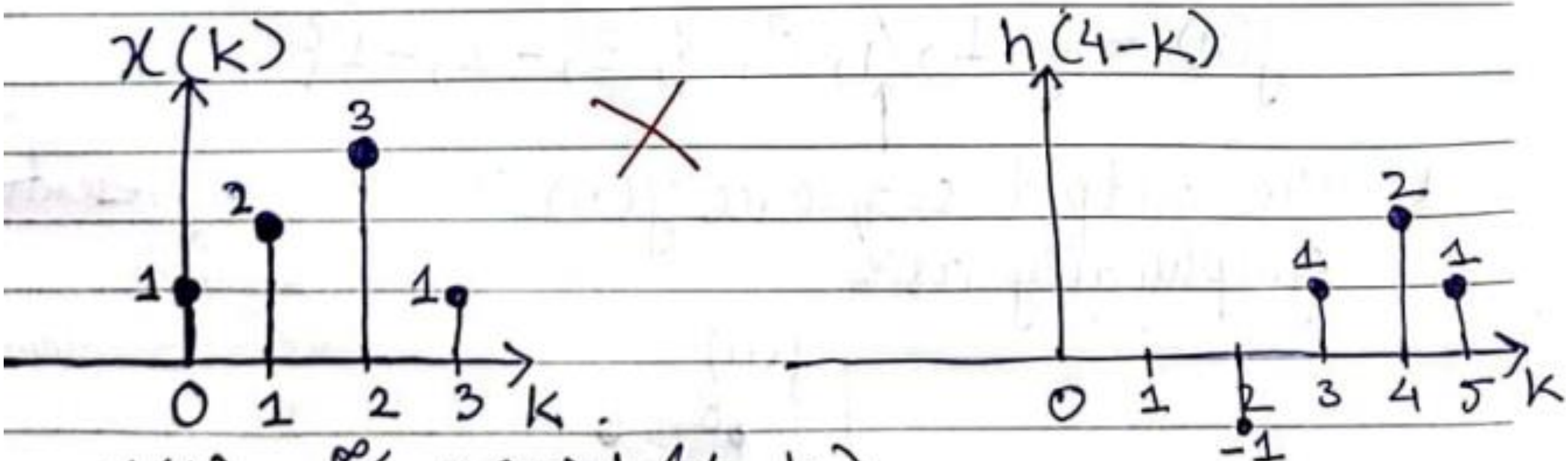
$$= (2)(-1) + (3)(1) + (1)(2)$$

$$= -2 + 3 + 2$$

$$y(3) = 3$$

# 5. Discrete Time Convolution

When $n = 4$:




$$y(4) = \sum_{k=-\infty}^{\infty} x(k)\,h(4-k)$$

$$= x(2)\cdot h(2) + x(3)\cdot h(3)$$

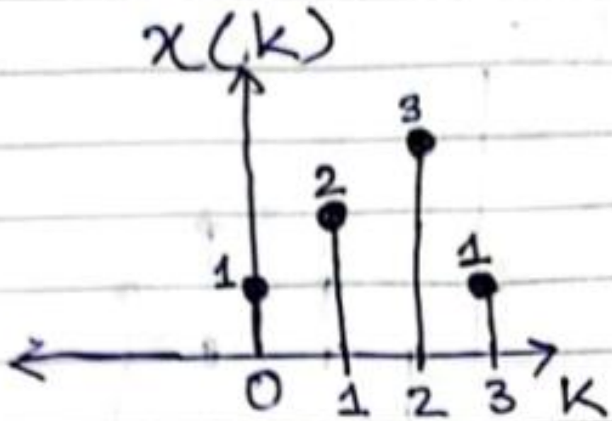
$$= (3)\cdot(-1) + (1)\cdot(1)$$

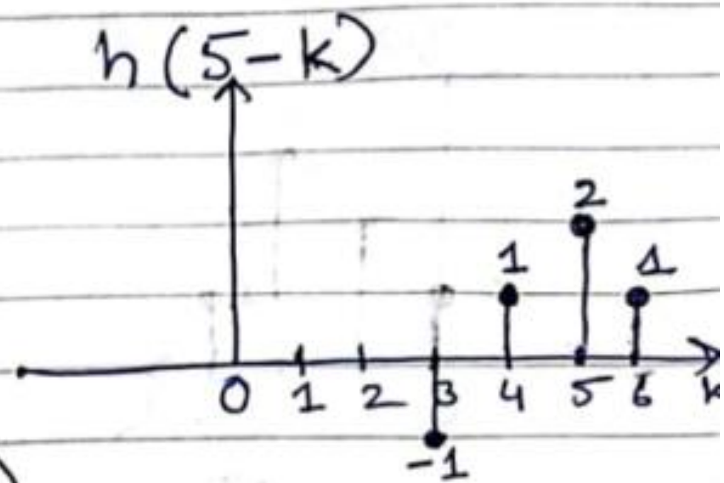
$$= -3 + 1$$

$$y(4) = -2$$

# 5. Discrete Time Convolution

when $n = 5$:




$$y(5) = \sum_{k=-\infty}^{\infty} x(k)\, h(5-k)$$

$$= x(3) \cdot h(3)$$

$$= (1)(-1)$$

$$y(5) = -1$$

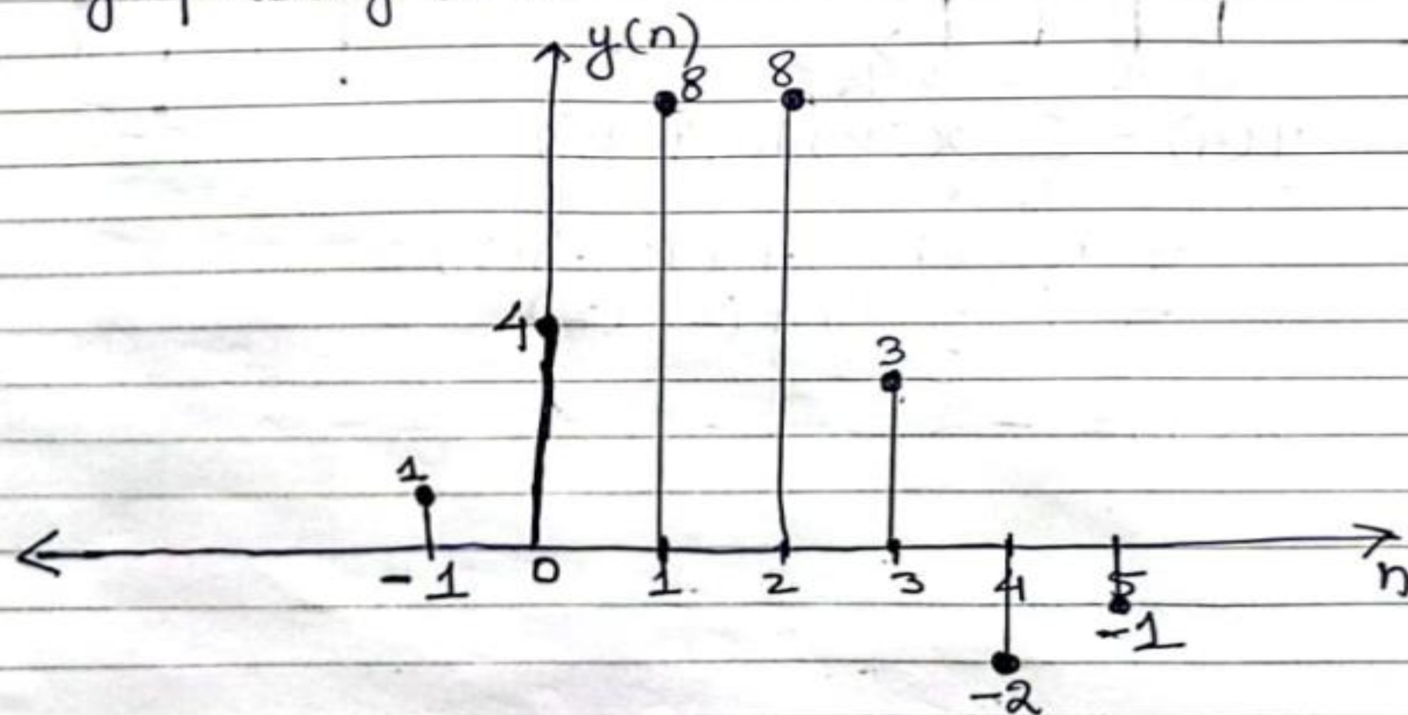
# 5. Discrete Time Convolution

* The output sequence y(n) can be written as:

$$y(n) = \{y(-1), y(0), y(1), y(2), y(3), y(4), y(5)\}$$

$$y(n) = \{1, \underset{\uparrow}{4}, 8, 8, 3, -2, -1\}$$

* The output sequence y(n) can be represented graphically as:



# 5. Discrete Time Convolution

## ii – Matrix Method

* The input sequence $x(n)$ is arranged as a column and the impulse response is arranged as a row as shown below. The elements of the two dimensional array are obtained by multiplying the corresponding row element with the column element. The sum of the diagonal elements gives the sample of $y(n)$.

$$\begin{array}{c|cccc}
x(n)\downarrow \;\backslash\; h(n)\rightarrow & 1 & 2 & 1 & -1 \\
\hline
1 & 1\times1 & 1\times2 & 1\times1 & 1\times(-1) \\
2 & 2\times1 & 2\times2 & 2\times1 & 2\times(-1) \\
3 & 3\times1 & 3\times2 & 3\times1 & 3\times(-1) \\
1 & 1\times1 & 1\times2 & 1\times1 & 1\times(-1)
\end{array}$$

# 5. Discrete Time Convolution

| $x(n) \downarrow$ \ $h(n) \rightarrow$ | 1 | 2 | 1 | -1 |
|---|---|---|---|---|
| 1 | 1 | 2 | 1 | -1 |
| 2 | 2 | 4 | 2 | -2 |
| 3 | 3 | 6 | 3 | -3 |
| 4 | 1 | 2 | 1 | -1 |

$y(-1) = 1$

$y(0) = 2+2 = 4$

$y(1) = 3+4+1 = 8$

$y(2) = 1+6+2(-1) = 8$

$y(3) = 2+3+(-2) = 3$

$y(4) = 1+(-3) = -2$

$y(5) = -1$

$y(n) = \{1, \underset{\uparrow}{4}, 8, 8, 3, -2, -1\}$

# 6. Properties of Convolution

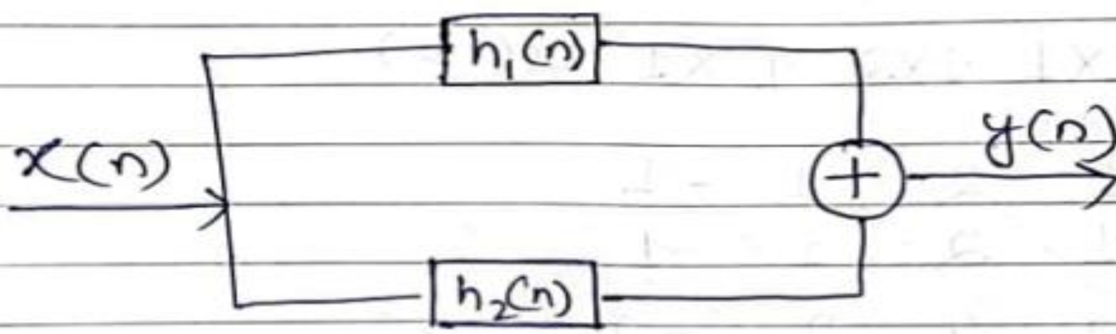
i) Commutative Property of Convolution

$$y(n) = \sum_{K=-\infty}^{\infty} x(K)\,h(n-K)$$

$$y(n) = \sum_{K=-\infty}^{\infty} h(k)\,x(n-K)$$

ii) Associative Property of Convolution


iii) Distributive Property of Convolution



# 7. Auto correlation and Cross correlation

## Home Work Examples

Determine the response of an LTI system whose impulse response $h(n)$ and input $x(n)$ are given by:

a) $h(n) = \{1, 2, \underset{\uparrow}{1}, -2, -1\}$

$x(n) = \{1, \underset{\uparrow}{2}, 3, -1, -3\}$

b) $h(n) = \begin{cases} 1 ; & 0 \leq n \leq 3 \\ 0 ; & n \geq 4 \end{cases}$

$x(n) = a^n u(n) ; \quad |a| < 1$

# 7. Auto correlation and Cross correlation

* the correlation of a sequence with itself is called auto correlation. i.e.,

$$r_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)\,x(n-m)$$

where $r_{xx}$ = the correlation sequence obtained by correlation of $x(n)$ with itself

$m$ = variable used for time shift.

* The correlation of a sequence $x(n)$ with $y(n)$ is called cross correlation.

"OR"

The correlation of two different sequences is called cross correlation

$$r_{xy}(m) = \sum_{n=-\infty}^{\infty} x(n)\,y(n-m)$$

# 7. Auto correlation and Cross correlation

**Example 2:** Perform cross correlation of the sequences.

$x(n) = \{1, 1, 2, 2\}$ and $y(n) = \{1, 3, 1\}$

**Solution:** [Using Matrix Method]

* $x(n)$ starts at $n_1 = 0$
* $y(n)$ starts at $n_2 = 0$
* Number of samples in $x(n) = N_1 = 4$
* Number of samples in $y(n) = N_2 = 3$
* Total number of samples in $r_{xy}(m) = N_1 + N_2 - 1$

$= 4 + 3 - 1$

$= 6$

* The sequence $r_{xy}(m)$ starts at i.e.,

$m_i = n_1 - (n_2 + N_2 - 1) = 0 - (0 + 3 - 1) = -2$

$m_i = -2$

# 7. Auto correlation and Cross correlation

* The sequence $r_{xy}(m)$ will finish at.

$$m_f = m_i + (N_1 + N_2 - 2)$$
$$= -2 + (4 + 3 - 2) = 3$$
$$m_f = 3$$

* For cross correlation input sequence $x(n)$ is arranged as a column and the output sequence $y(n)$ is arranged as row.
* In row we will place $y(-n)$ [folding] instead of $y(n)$ Therefore

# 7. Auto correlation and Cross correlation

$y(n) = \{1, 3, 1\}$

$y(-n) = \{1, 3, \underset{\uparrow}{1}\}$

| $x(n) \downarrow$ \ $y(-n) \rightarrow$ | 1 | 3 | 1 |
|---|---|---|---|
| 1 | 1 | 3 | 1 |
| 1 | 1 | 3 | 1 |
| 2 | 2 | 6 | 2 |
| 2 | 2 | 6 | 2 |

# 7. Auto correlation and Cross correlation

$r_{xy}(-2) = 1$

$r_{xy}(-1) = 1+3 = 4$

$r_{xy}(0) = 2+3+1 = 6$

$r_{xy}(1) = 2+6+1 = 9$

$r_{xy}(2) = 6+2 = 8$

$r_{xy}(3) = 2$

$r_{xy}(m) = \{1, 4, \underset{\uparrow}{6}, 9, 8, 2\}$

# 7. Auto correlation and Cross correlation

Home Work

Perform cross correlation of the sequences given in Example 2 using Graphical Method.

Example 3: Determine the autocorrelation sequence for $x(n) = \{1, 2, 3, 4\}$.

Solution: [Using Matrix Method]

Let $r_{xx}(m)$ be the auto correlation sequence.

The autocorrelation sequence $r_{xx}(m)$ is given by

$$r_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)\,x(n-m)$$

# 7. Auto correlation and Cross correlation

* For auto correlation input sequence $x(n)$ is arranged as a column and the ... sequence $x(-n)$ is arranged as row.

$$x(n) = \{1, 2, 3, 4\}$$

$$x(-n) = \{4, 3, 2, 1\}$$

* Total Number of samples in $r_{xx}(m)$ will be

$$N+N-1 = 4+4-1 = 7$$

* The first sample of $r_{xx}(m)$ will start at

$$m_i = n-(n+N-1) = 0-(0+4-1) = -3$$

$$m_i = -3$$

* The last sample of $r_{xx}(m)$ will be at

$$m_f = m_i+(N+N-2) = -3+(4+4-2) = 3$$

# 7. Auto correlation and Cross correlation

| $x(n) \downarrow$ \ $x(-n) \rightarrow$ | 4 | 3 | 2 | 1 |
|---|---|---|---|---|
| 1 | 4 | 3 | 2 | 1 |
| 2 | 8 | 6 | 4 | 2 |
| 3 | 12 | 9 | 6 | 3 |
| 4 | 16 | 12 | 8 | 4 |

$r_{xx}(-3) = 4$

$r_{xx}(-2) = 8+3 = 11$

$r_{xx}(-1) = 12+6+2 = 20$

$r_{xx}(0) = 16+9+4+1 = 30$

$r_{xx}(1) = 12+6+2 = 20$

$r_{xx}(2) = 8+3 = 11$

$r_{xx}(3) = 4$

$r_{xx}(m) = \{4, 11, 20, \underset{\uparrow}{30}, 20, 11, 4\}$

# 7. Auto correlation and Cross correlation

**Home Work**

Perform Autocorrelation on given sequence using Graphical Method.

$$x(n) = \{1, \underset{\uparrow}{4}, -3, 2, 1\}$$

## References: Textbooks


1. *"Discrete-Time Signal Processing", International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *"Digital Signal Processing Using Matlab", 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*